Effects of plant-based diets on plasma lipids.

Dyslipidemia is a primary risk factor for cardiovascular disease, peripheral vascular disease, and stroke. Current guidelines recommend diet as first-line therapy for patients with elevated plasma cholesterol concentrations. However, what constitutes an optimal dietary regimen remains a matter of controversy. Large prospective trials have demonstrated that populations following plant-based diets, particularly vegetarian and vegan diets, are at lower risk for ischemic heart disease mortality. The investigators therefore reviewed the published scientific research to determine the effectiveness of plant-based diets in modifying plasma lipid concentrations. Twenty-seven randomized controlled and observational trials were included. Of the 4 types of plant-based diets considered, interventions testing a combination diet (a vegetarian or vegan diet combined with nuts, soy, and/or fiber) demonstrated the greatest effects (up to 35% plasma low-density lipoprotein cholesterol reduction), followed by vegan and ovolactovegetarian diets. Interventions allowing small amounts of lean meat demonstrated less dramatic reductions in total cholesterol and low-density lipoprotein levels. In conclusion, plant-based dietary interventions are effective in lowering plasma cholesterol concentrations.